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Composite cores and tamper yield: Lesser-known aspects of Manhattan Project fission bombs

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This paper examines the physics of two lesser-known aspects of the nuclear weapons produced in the Manhattan Project: The idea of constructing weapons with so-called composite cores (that is, ones which would have comprised combinations of uranium and plutonium), and the fact that some 30% of the plutonium-fueled *Fat Man* yield was due to fissions in the uranium-238 tamper of that weapon, fissions induced by high-energy neutrons liberated by fissions in the plutonium core. Simple models of both issues give results in reasonable accord with available data.
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I. INTRODUCTION

This paper continues a series of papers published in this journal on the physics of Manhattan Project fission weapons; see Ref. 1 and references therein. This contribution addresses two lesser-known aspects of these weapons. The first of these is the notion of constructing weapons with *composite* cores, that is, ones with cores comprising both uranium-235 (^{235}U) and plutonium-239 (^{239}Pu). The Manhattan Engineer District's uranium-enrichment and plutonium-synthesis facilities produced material at quite different rates; by combining the two materials with masses dictated by the ratio of production rates, the rate of bomb production could be maximized. In the immediate aftermath of the Hiroshima and Nagasaki missions, it was not clear when the Japanese might surrender and how many more bombs might be needed; rapid production might have proven essential to ending the war. This was especially so because, at that time, Army Chief of Staff General George Marshall was seriously considering reserving future bombs for tactical use during an invasion of Japan, which was scheduled to begin on November 1, 1945.[2,3] With the successful *Trinity* test of the very efficient implosion design (using only plutonium), most bombs following the Nagasaki mission were expected to be of the implosion type. Implosion had to be used with plutonium bombs to overcome that isotope's propensity to pre-detonate; ^{235}U suffered no such problem, so it could be used in either the less efficient Hiroshima gun-type *Little Boy* design or the *Fat Man* implosion design. The idea of constructing bombs with composite cores was raised immediately following the *Trinity* test, although they were not put into use at the time. The first composite-core weapon was tested in Operation Sandstone in 1948, but my concern in this paper is with the situation in mid-1945. I examine fissile-material production rates in Sec. II, and the composite-core issue proper in Sec. III by making some estimates of threshold-critical core masses for plausible composite configurations. The physics calculations underlying these estimates are gathered in the Appendix so as not to disrupt the flow of the description.

The second issue addressed in this paper concerns an aspect of the *Trinity* and *Fat Man* implosion bombs. Radiochemical and spectroscopic analyses of fallout products trapped in fused desert sand ("Trinitite") after the test have revealed that ~31% ± 2% of the yield of the explosion resulted from fissions induced in uranium-238 (^{238}U) in the tamper of this weapon by high-energy neutrons which escaped from the exploding plutonium core.[4] Section IV examines the ^{238}U yield with a model involving fundamental energetics and reaction-cross-sections. Section V offers a few concluding remarks.

It is worthwhile addressing a question I am often asked by students: Exactly how many bombs were expected to be available for use through the remainder of 1945? After the Nagasaki mission, another implosion bomb was expected to be ready in about another week, although the fissile material for this bomb never left the United States. The schedule for the remainder of the year was outlined by Manhattan commander General Leslie Groves in a July 30, 1945, memo to General Marshall; Groves also touched on the composite-core issue:[5]

> *In September, we should have three or four bombs. One of these will be made from 235 material and will have a smaller effectiveness, about two-thirds that of the test type, but by November, we should be able to bring this up to full power. There should be either four or three bombs in October, one of the lesser size. In November there should be at least five bombs and the rate will rise to seven in December and increase decidedly in early 1946. By some time in November, we should have the effectiveness of the 235 implosion type bomb equal to that of the tested plutonium implosion type.*
>
> *By mid-October we could increase the number of bombs slightly by changing our design … to one using both materials in the same bomb. I have not made this change because of the ever present possibilities of difficulties in new designs …*

Thus, some 18–20 bombs were expected to be available from September through the end of the year, or about one every six to seven days. This rapid pace testifies to the fact that all fissile-material production plants were reaching full capacity at that time. The idea of composite cores had been put to Groves by Robert Oppenheimer on July 19, just three days after the *Trinity* test. Groves decided to stick with existing plans, replying that[6]

> *Factors beyond our control prevent us from considering any decision, other than to proceed according to the existing schedule for the time*

being. It is necessary to drop the first Little Boy and the first Fat Man and probably a second one in accordance with our original plans. It may be that as many as three of the latter in their best present form may have to be dropped to conform to planned strategic operations.

Even before Hiroshima and Nagasaki, Los Alamos scientists were thinking about improvements in bomb design.

II. FISSILE MATERIAL PRODUCTION RATES

As described in a previous paper in this series, the assembled Hiroshima *Little Boy* bomb core comprised a cylinder of highly enriched ^{235}U of mass $\sim$66 kg surrounded by a cylindrical tungsten-carbide (WC; steel) tamper of mass $\sim$550 kg.[7] As described below, this corresponds to about four tamped critical masses of ^{235}U for this tamper material and mass.

Production of ^{235}U at Oak Ridge, Tennessee, ramped up steadily during 1945 as various enrichment facilities (liquid and gaseous diffusion; electromagnetic separation) came into full operation. According to figures given in the official chronicle of the project, the *Manhattan District History*, 93.9 kg of ^{235}U (85%–90% enrichment) were produced between September 9 and December 2, 1945, or about 1.12 kg per day.[8] Between December 2 and December 30, the rate accelerated to about 1.7 kg per day, an improvement reflected in the estimates given in Groves's memo. For sake of computational convenience, I will assume a production rate of 1.2 kg per day for the immediate post-war period; at this rate a 66 kg core would require just under eight weeks (55 days, to be exact) of production time.

The *Fat Man* bomb had a spherical configuration due to its implosion design, and comprised a core of ^{239}Pu of mass $\sim$6.3 kg surrounded by nested tamper spheres of depleted uranium (DU; essentially pure ^{238}U) and aluminum totaling $\sim$230 kg. The compressed tamped critical mass of ^{239}Pu for this configuration is about 1.44 kg, so we again have $\sim$ four tamped critical masses for the operational bomb. The rate of production of plutonium in a reactor fueled with natural uranium is about 0.75 g per day per megawatt of operating power. The Manhattan Project's three reactors located at Hanford, Washington, each operated at $\sim$250 MW; all had come on-line by March, 1945. This corresponds to $\sim$560 g of Pu per day (assuming high chemical separation efficiency), or just over 11 days between each 6.3 kg core. Five such cores would require just over 56 days, the same duration as for a single *Little Boy* core estimated above. Overall, these figures predict six bombs (five *Fat Man* models and one *Little Boy*) about every eight weeks in the immediate post-Hiroshima period; this is in sensible agreement with Groves' estimate of 6–8 bombs over the months of September and October. ^{235}U was produced at about twice the rate of ^{239}Pu, but the plutonium bomb used much less material in view of the greater inherent fissility of that element (larger fission cross-section; more neutrons per fission) and the efficiency of the implosion design, which achieved a compression ratio estimated to be as high as $\sim$2.5.[4]

The above figures indicate that an optimal composite-core configuration would contain ^{235}U and ^{239}Pu in a mass ratio of about 2:1. The key physics questions are then: How does the critical mass of a (^{235}U + ^{239}Pu) composite core, under, say, a compression ratio of 2.5, compare to that for a ^{239}Pu-only bomb? Also, if four critical masses are to be used per bomb, at what rate can bombs be produced? These questions are taken up in Sec. III.

III. COMPOSITE-CORE CRITICALITY

A simplified neutron-diffusion model of tamped criticality was developed in another paper in this series.[9] For the present purpose, I have used the approach described therein, modified to modeling a composite core as comprising an inner core of fissile material of outer radius R_1 and a surrounding outer core of fissile material of outer radius R_2, all of which is surrounded by a tamper of outer radius R_{tamp}. This idealized model is sketched in Fig. 1. I simplify the analysis by assuming that the tamper comprises a single non-fissile shell of ^{238}U of total mass 230 kg.

The physics of the threshold-criticality calculation, in which the neutron density in the core/tamper assembly is neither growing nor diminishing with time, is detailed in the Appendix. Here, I discuss the resulting numerical values; all parameter values for fissile and tamper materials were taken to be those in Tables I and II of Ref. 7.

Results are summarized in Table I of the current paper; the essential result, the total critical mass, appears in the third column. The first two scenarios are baseline "bare" (untamped), uncompressed threshold critical masses for ^{235}U and ^{239}Pu. Scenario 3 corresponds to an idealized spherical *Little Boy* device with an uncompressed 550 kg WC tamper; this gives the source of the four critical-massses ($\sim$66 kg/16.2 kg) cited above. Similarly, scenario 4, an idealized *Fat Man* with a 230 kg DU tamper and compression ratio 2.5, indicates the similar four-critical-mass configuration for that weapon, $\sim$6.3 kg/1.44 kg $\sim$4.4. Scenario 5 substitutes a ^{235}U core in the implosion model; the dramatic decrease in critical mass facilitated by compression (compare to scenario 3) is evident.

The analysis presented in the Appendix results in an expression that must be satisfied to achieve threshold criticality. The only adjustable parameters aside from the compression ratio are the radii R_1, R_2, and R_{tamp}. To examine composite-core configurations, my procedure was to fix the tamper mass as 230 kg DU, choose a mass for the inner core, and then use the goal-seek function in a spreadsheet to determine values of R_2

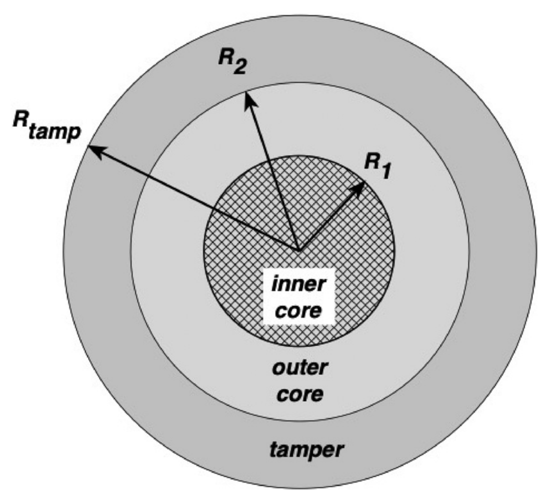

Fig. 1. *Conceptual* sketch of composite core with tamper. The inner and outer cores are presumed to be made of fissile material; the tamper is non-fissile. This sketch is purely schematic and not to scale.

Table I. Threshold-critical masses for various ^{235}U/^{239}Pu configurations.

Scenario	Core material (inner + outer)	M_{crit} (kg)	Tamper	Compression ratio	Comments
1	^{235}U	45.9	None	1	Bare ^{235}U
2	^{239}Pu	16.7	None	1	Bare ^{239}Pu
3	^{235}U	16.2	550 kg WC	1	Idealized *Little Boy*[a]
4	^{239}Pu	1.44	230 kg DU	2.5	Idealized *Fat Man*[b]
5	^{235}U	3.5	230 kg DU	2.5	Pure ^{235}U implosion core
6	^{235}U + ^{239}Pu	1.8 + 0.9	230 kg DU	2.5	Composite core with 2:1 ^{235}U: ^{239}Pu mass ratio
7	^{239}Pu + ^{235}U	0.7 + 1.4	230 kg DU	2.5	Composite core with 2:1 ^{235}U: ^{239}Pu mass ratio

[a]Assumes spherical core/tamper; real *Little Boy* had a cylindrical core/tamper. WC = tungsten-carbide.
[b]Assumes single spherical tamper of depleted uranium (DU = ^{238}U) in place of true DU + aluminum tamper spheres. Compression ratio that of *Trinity* test (2.5).

and R_{tamp} which satisfy the criticality constraint. In this scheme, the only real unknown is R_2, because R_{tamp} is ultimately dictated by the adopted tamper mass. I then adjusted my choice of the mass of the inner core until the ^{235}U: ^{239}Pu mass ratio of the solution was 2:1, as discussed above. The results are summarized in scenarios 6 and 7 in Table I. Because ^{235}U and ^{239}Pu have different fission properties, the results depend on which is used for the inner/outer core.

The best compressed composite design would be scenario 7, which has ^{239}Pu as its inner core; the total mass is about 2.1 kg (more precisely, the masses are ~0.71 and 1.43 kg). This total is greater than that of a ^{239}Pu-only core (scenario 4), but can be produced more quickly. If we again take four critical masses for an operational bomb, production of 2.8 kg of ^{239}Pu will require a little over five days at ~0.56 kg per day (or, similarly, ~5.7 kg of ^{235}U at 1.2 kg per day). Two months would then see production of 10–11 bombs, almost double the total of six estimated to be available at the start of this section on the basis of sticking with the original *Little Boy* and *Fat Man* designs.

Given that ^{235}U is less fissile than ^{239}Pu, how might the yield of an imploded composite-core weapon compare to the actual *Fat Man* implosion device used at Nagasaki? An accurate prediction would require a detailed time-dependent simulation, but a rough estimate can be made with the "toy" yield model of Ref. 7. Running this model for an 8 kg ^{239}Pu core plus 230 kg DU tamper and compression ratio of 2.5 gives an estimated yield of ~28.0 kilotons (kt); I used 8 kg as it represents four composite-core critical masses (scenario 7 in Table I), approximately the ratio of actual to critical masses used in both *Little Boy* and *Fat Man*. An 8 kg ^{235}U core with the same tamper and compression is estimated to yield ~5.5 kt. Weighting these in a 1:2 proportion gives a yield of ~13 kt. The real *Fat Man* achieved a yield of about 21 kt, but at the expense of over twice as much ^{239}Pu. Lower-yield bombs might in fact have been considered an advantage in tactical-use scenarios.

An additional advantage of a composite core is that it would involve a much reduced spontaneous-fission background compared with a ^{239}Pu-only core because of the much smaller amount of that material involved.

IV. FAT MAN ^{238}U TAMPER YIELD

As described in the introduction, some 30% of the yield of the *Trinity* test resulted from fissions induced in the ^{238}U tamper sphere of that device which surrounded the ^{239}Pu core. Given that ^{238}U is well-known to be non-fissile, this result may seem surprising.

A detailed analysis of the physics of this figure would be extremely difficult. Fissions which occur in either the core or the tamper could be caused by neutrons that were originally emitted in either material and subsequently (i) stayed in their birth material and caused a fission, possibly after scattering several times; (ii) returned to their birth material after one or more escape/reflection trips into and out of the other material; or (iii) caused a fission in the other material, again possibly after several trips into and out of each. Here I develop a pedagogical-level explanation of the ^{238}U yield fraction in a way that attempts to incorporate the various possible neutron fates in a statistically and energetically plausible way without getting tangled up in trying to track multiple generations of neutron creation and travel. To do so, I simplify the situation by assuming that any neutrons which migrate from the core into the tamper remain within the tamper, and that any fissions induced within the tamper arise only from core-escaped neutrons and not from secondary neutrons released in fissions within the tamper. On this basis, I can set up a four-factor formula to model the ratio of ^{238}U to ^{239}Pu fissions

$$\frac{^{238}\text{U fissions}}{^{239}\text{Pu fissions}} \sim \frac{0.3}{0.7} \sim 0.43. \tag{1}$$

Each factor can be estimated from straightforward energy and cross-section considerations, as follows.

Assume that a total of F fissions occur in the plutonium core; the intent here is that this number incorporates fissions due to neutrons arising from anywhere within the core/tamper structure. If the number of secondary neutrons liberated per fission is ν, then the net number of neutrons liberated will be $F(\nu - 1)$; the "–1" appears because one neutron is consumed in causing each fission.

Some of these secondary neutrons will escape from the core into the surrounding tamper. If the probability of escape is p_{esc}, then $F(\nu - 1)\, p_{esc}$ neutrons will enter the tamper. However, only those neutrons which are more energetic than the fission threshold of ^{238}U will have a chance of inducing a fission. Designate the fraction of neutrons that are sufficiently energetic as f_{thresh}. This will give $F(\nu - 1)\, p_{esc}\, f_{thresh}$ neutrons than can potentially induce fissions in the tamper, but not all will do so because of competing effects, notably inelastic collisions with ^{238}U nuclei. If the probability that a neutron will induce a fission is p_{fiss}, then the four-factor analog of Eq. (1) is

$$\frac{^{238}\text{U fissions}}{^{239}\text{Pu fissions}} = \frac{F(\nu-1)p_{\text{esc}}f_{\text{thresh}}p_{\text{fiss}}}{F} = (\nu-1)p_{\text{esc}}f_{\text{thresh}}p_{\text{fiss}}. \quad (2)$$

Each of these factors is analyzed and estimated in the following paragraphs:

(i) *Secondary neutrons*. This is the most straightforward factor. As described in Ref. 1 and references therein, $\nu \sim 3.2$ for ^{239}Pu, so $(\nu - 1) \sim 2.2$.

(ii) *Escape probability*. The probability that a neutron liberated somewhere within a sphere of radius R will reach the surface of the sphere and escape without being consumed in a fission was analyzed in another paper in this series.[10] This analysis was predicated on extending the usual exponential law for the probability of a neutron to pass through a material of linear distance L, $p = \exp(-\sigma_{\text{total}}\, nL)$, to apply to a sphere by integrating over all directions of travel within the sphere from every possible point of emission out to the edge of the sphere. σ_{total} is the sum of all possible reaction cross sections; I take the relevant possibilities to be fission, elastic scattering, and inelastic scattering. n is the number density of nuclei in the material. As shown in Ref. 10, the escape probability depends on the maximum number of scatterings S that a neutron can suffer before it escapes, and can be written as

$$p_{\text{esc}} = P_{\text{sph}}\left[\frac{1-g^{S+1}}{1-g}\right], \quad (3)$$

where

$$g = \left(\frac{\sigma_{\text{scatter}}}{\sigma_{\text{total}}}\right)(1-P_{\text{sph}}). \quad (4)$$

Here, σ_{scatter} is the sum of the cross-sections for inelastic and elastic scattering and P_{sph} is the probability that a neutron will escape directly from within the sphere without experiencing any interaction. In Ref. 10, an integral expression for P_{sph} was developed, which was computed numerically. However, it can be shown that the integral can be solved analytically, and reduces to[11]

$$P_{\text{sph}} = \frac{3}{8x^3}\left\{2x^2 + e^{-2x}(2x+1) - 1\right\}, \quad (5)$$

where

$$x = \sigma_{\text{total}}\, nR, \quad (6)$$

in which R is the radius of the core. If the core has been compressed, this can be accommodated by adjusting the number density n appropriately.

I adopt a compression ratio of 2.5 to be consistent with the discussion of composite cores in Sec. III. For ^{239}Pu, a 6.3 kg core compressed by this factor will have a radius $R \sim 3.38$ cm, and a number density $n = 9.82 \times 10^{28}\ \text{m}^{-3}$. (For sake of setting up a simple estimate, I ignore the effect of the polonium-beryllium initiator that was placed at the center of the core.) The question of how many scatterings to allow is difficult to pin down. A nuclear explosion can involve a few dozen fission generations, so a value of S of at least a few should be reasonable. For the 6.3 kg core numbers given above, $p_{\text{esc}} = (0.273, 0.425, 0.509, 0.556, 0.583, 0.597)$ for $S = (0, 1, 2, 3, 4, 5)$. For larger values of S, p_{esc} does not change drastically; $p_{\text{esc}} \to \sim 0.61$ for $S \to \infty$. For estimating the ^{238}U yield, I adopt $p_{\text{esc}} \sim 0.55$, which corresponds to $S \sim 3$.

(iii) *Neutrons above the fission threshold energy*. Neutrons emitted in fissions emerge with a spectrum of kinetic energies up to several MeV, with an average of about 2 MeV. The energy spectrum can be empirically fit with a Maxwell-type distribution which incorporates a fitting parameter akin to a temperature.[12] What is of interest here is the fraction of neutrons that have a kinetic energy above some threshold energy E, which is the difference between the energy liberated upon neutron capture and the "fission barrier." With such a distribution, this is given by

$$f_{\text{thresh}} = \frac{2}{\sqrt{\pi}}\int_{E/\alpha}^{\infty}\sqrt{x}e^{-x}dx, \quad (7)$$

where α is the temperature-like parameter. For ^{235}U, $\alpha \sim 1.29$ MeV, which I assume also applies to ^{239}Pu; experimentally, the energy spectrum of neutrons emitted by the two under thermal-neutron bombardment are statistically very similar.[13] This integral can be expressed in a form suited to spreadsheet computation by a transformation which renders it in terms of the "error function" *erf* of statistics

$$f_{\text{thresh}} = 1 + \frac{2}{\sqrt{\pi}}z\,e^{-z^2} - erf(z), \quad (8)$$

where

$$z = \sqrt{E/\alpha}. \quad (9)$$

Because fission is such a complex phenomenon, barrier energies are difficult to determine either experimentally or theoretically. A recent experimental determination for that of ^{239}U (the compound nucleus formed when ^{238}U captures a neutron) gives $\sim$6.3 MeV.[14] In a recent paper, Möller *et al.* presented a tabulation of *calculated* fission barriers for over 5000 heavy nuclides with atomic numbers from 51 to 130 based on a model which incorporates five nuclear-shape degrees of freedom; altogether, they modeled over five million deformation shapes.[15] For ^{239}U they estimated $\sim$6.2 MeV, close to the experimental value, but cautioned that their estimated barriers agree with known data to within only $\sim\pm 1$ MeV. When ^{238}U captures a neutron, about 4.8 MeV of binding energy is released, so these results indicate that only neutrons with energies of $\sim$1.4–1.5 MeV or greater have a chance of inducing a fission in that isotope. With these values, Eqs. (8) and (9) give $z \sim 1.04$–1.08 and $f_{\text{thresh}} \sim 0.54$–$0.51$. I adopt $f_{\text{thresh}} \sim 0.5$.

(iv) *Probability of fission*. This is the most difficult of the four parameters in Eq. (2) to estimate with any precision. Neutrons within the ^{238}U tamper which suffer elastic collisions will not lose any kinetic energy and so can go on to possibly induce a fission, whereas those which suffer inelastic collisions tend to lose so much energy that they fall below the ~1.4-MeV threshold discussed above. As a simple approach to estimating $p_{\rm fiss}$, I presume that neutrons which find themselves within the tamper do not escape from the tamper (this is discussed further below), and take the probability of causing a fission to be the ratio of the sum of the fission and elastic-scattering cross-sections to the total cross-section

$$p_{\rm fiss} \sim \frac{\sigma_{\rm fiss} + \sigma_{\rm elastic}}{\sigma_{\rm total}} \sim \frac{5.112 \text{ bn}}{7.707 \text{ bn}} \sim 0.66. \tag{10}$$

The assumption that neutrons which enter the tamper remain within it until causing a fission or being casualties of inelastic collisions can be given a rough justification as follows. Again neglecting the neutron initiator located within the core, a 100 kg ^{238}U tamper will have an outer radius of about 8.2 cm for a compression factor of 2.5 and a Pu-core radius of 3.4 cm. (The 100 kg used here is the mass of *Fat Man's* ^{238}U tamper; the 230 kg used in the composite-core calculations is the total mass of the uranium and aluminum tamper spheres.) This will give the tamper shell a (compressed) thickness of ~4.8 cm. For this compression ratio, the neutron mean free path $\lambda \sim 1/(\sigma_{\rm total}\, n)$ is ~1.2 cm, about one-quarter of the shell thickness: Most neutrons will thus likely interact with nuclei within the shell. In the *Fat Man* bombs, the ^{238}U tamper shell was surrounded by an aluminum shell, so many neutrons that might otherwise have escaped from the ^{238}U tamper will have been reflected back into it.

Gathering the results of the preceding paragraphs gives

$$\frac{^{238}\text{U fissions}}{^{239}\text{Pu fissions}} = (\nu - 1) p_{\rm esc} f_{\rm thresh}\, p_{\rm fiss} \sim (2.2)(0.55)(0.5)(0.66) \sim 0.40. \tag{11}$$

This result is in surprisingly good agreement with the value suggested by the Trinitite data as estimated in Eq. (1); the analysis must be the beneficiary of multiple cancelling approximations which mask the complexities of dealing with the fates of multiple generations of neutrons. The escape probability might well be argued to be somewhat higher than has been estimated by virtue of allowing more scatterings or adopting a slightly lower compression ratio; conversely, the adopted fission probability is likely optimistic because neutron loss from the tamper has been neglected. The threshold fraction could go either way depending on the actual value of the fission barrier for ^{239}U, but is probably not far wrong. While it is satisfying to see that a simple estimate predicts a result solidly in line with the experimental evidence, this must admittedly be regarded as a zeroth-order model which begs for a more substantial analysis.

From a hydrodynamic perspective, nesting *Fat Man*'s ^{238}U tamper shell within an aluminum tamper shell was challenging in that it was known that accelerating a light metal into a heavier one could lead to so called Rayleigh-Taylor instabilities wherein "jets" of the light metal could race ahead of the main body of that material unless the implosion was very symmetric. But arranging the tampers in this way significantly enhanced the yield of the plutonium bomb, which otherwise would not have been much greater than that of the *Little Boy* design. Later-generation thermonuclear weapons also achieved enhanced yields by surrounding their fission-fusion cores with casings of ^{238}U to achieve a three-stage "fission-fusion-fission" explosion.

V. CONCLUDING REMARKS

The essential conclusions of this paper are that composite-core designs would have offered the architects of the Manhattan Project a significant increase in bomb availability without a drastic sacrifice of yield, and that using a tamper material which itself had some fast-neutron fissility could significantly boost bomb yield.

These considerations were but just the first steps in making nuclear weapons more efficient. The immense postwar proliferation in bomb designs configured to address a multitude of missions had its origins in the summer of 1945, weeks before the world learned of Hiroshima and Nagasaki.

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APPENDIX: COMPOSITE CORE + TAMPER THRESHOLD CRITICALITY

For a primer on the physics of neutron diffusion, consult Ref. 9 and Chapter 2 of Ref. 16. This Appendix is intended only as a summary of essential relationships, not as a source of derivations.

In determining critical masses, the key quantity is the neutron density N within the core or tamper; do not confuse N and the number density of nuclei n. Threshold criticality occurs when the neutron density is neither increasing nor decreasing with time. In this case, diffusion theory provides the following solutions for the neutron density as a function of the spherical coordinate r as measured from the center of the inner core of Fig. 1 of the text

$$N_1(r) = A\frac{d_1}{r}\sin\left(\frac{r}{d_1}\right), \tag{A1}$$

$$N_2(r) = B\frac{d_2}{r}\sin\left(\frac{r}{d_2}\right) + C\frac{d_2}{r}\cos\left(\frac{r}{d_2}\right), \tag{A2}$$

and

$$N_{\rm tamp}(r) = \frac{D}{r} + E. \tag{A3}$$

Subscripts 1, 2, and tamp will be used to designate the inner core, outer core, and tamper, respectively. (A, B, C, D, E) are constants of integration to be eliminated upon application of boundary conditions as discussed below. The solution for the outer core, Eq. (A2), contains a cosine term, which does not appear in the solution for the inner core, Eq. (A1). Equation (A2) is the general solution of the time-independent diffusion equation for a fissile material, but the cosine term is dropped in Eq. (A1) as it would diverge at $r = 0$. d_1 and d_2 are characteristic size scales of the inner and outer cores given by

$$d_1 = \sqrt{\frac{\lambda_1^{\rm fiss}\lambda_1^{\rm tran}}{3(\nu_1 - 1)}}, \tag{A4}$$

and

$$d_2 = \sqrt{\frac{\lambda_2^{\rm fiss}\lambda_2^{\rm tran}}{3(\nu_2 - 1)}}. \tag{A5}$$

$\lambda^{\rm fiss}$ and $\lambda^{\rm tran}$ are, respectively, the mean-free-paths for neutrons for fission and transport. These are given by nuclear number densities and cross-sections according as

$$\lambda^{\rm tran} = \frac{1}{n(\sigma^{\rm fiss} + \sigma^{\rm elas})}, \tag{A6}$$

and

$$\lambda^{\rm fiss} = \frac{1}{n\sigma^{\rm fiss}}, \tag{A7}$$

where $\sigma^{\rm fiss}$ is the fission cross-section and $\sigma^{\rm elas}$ the elastic-scattering cross-section for the inner or outer core material as appropriate. ν_1 and ν_2 in Eqs. (A4) and (A5) are the number of neutrons liberated per fission for the two core materials.

Five boundary conditions dictate the continuity of neutron density and flux between the inner and outer cores, the outer core and the tamper, and at the outer edge of the tamper. These are

$$N_1(R_1) = N_2(R_1), \tag{A8}$$

$$\lambda_1^{\rm tran}\left(\frac{\partial N_1}{\partial r}\right)_{R_1} = \lambda_2^{\rm tran}\left(\frac{\partial N_2}{\partial r}\right)_{R_1}, \tag{A9}$$

$$N_2(R_2) = N_{\rm tamp}(R_2), \tag{A10}$$

$$\lambda_2^{\rm tran}\left(\frac{\partial N_2}{\partial r}\right)_{R_2} = \lambda_{\rm tamp}^{\rm tran}\left(\frac{\partial N_{\rm tamp}}{\partial r}\right)_{R_2}, \tag{A11}$$

and

$$N_{\rm tamp}(R_{\rm tamp}) = -\frac{2}{3}\lambda_{\rm tamp}^{\rm tran}\left(\frac{\partial N_{\rm tamp}}{\partial r}\right)_{R_{\rm tamp}}. \tag{A12}$$

With these five conditions, the constants (A, B, C, D, E) in Eqs. (A1)–(A3) can be eliminated. The algebra is lengthy; the result is an expression which, aside from nuclear factors such as values of σ and n, involves only the three radii (R_1, R_2, $R_{\rm tamp}$). This can be written as

$$U - QV = 0. \tag{A13}$$

U, Q, and V are

$$U = \left(\frac{B}{C}\right)\frac{\sin x_3}{x_3} + \frac{\cos x_3}{x_3}, \tag{A14}$$

$$Q = \psi\left\{\left(\frac{B}{C}\right)[-\sin x_3 + x_3\cos x_3] - [\cos x_3 + x_3 \sin x_3]\right\}, \tag{A15}$$

and

$$V = \frac{2\lambda_{\rm tamp}^{\rm tran}}{3R_{\rm tamp}^2} - \frac{1}{R_{\rm tamp}} + \frac{1}{R_2}. \tag{A16}$$

The various quantities appearing in these expressions are

$$(x_1,\ x_2,\ x_3) = \left(\frac{R_1}{d_1},\ \frac{R_1}{d_2},\ \frac{R_2}{d_2}\right), \tag{A17}$$

$$\left(\frac{B}{C}\right) = \frac{[\delta Y \sin x_1 + Z\cos x_2]}{[\delta W \sin x_1 - Z \sin x_2]}, \tag{A18}$$

and

$$\psi = -\frac{\lambda_2^{\rm tran} d_2}{\lambda_{\rm tamp}^{\rm tran}}, \tag{A19}$$

with

$$W = [-\sin x_2 + x_2 \cos x_2], \tag{A20}$$

$$Y = [\cos x_2 + x_2 \sin x_2], \tag{A21}$$

$$Z = \lambda[-\sin x_1 + x_1 \cos x_1], \tag{A22}$$

where

$$\lambda = \frac{\lambda_1^{\rm tran}}{\lambda_2^{\rm tran}}\frac{d_1}{d_2} = \frac{\lambda_1^{\rm tran}}{\lambda_2^{\rm tran}}\delta. \tag{A23}$$

For δ in Eq. (A18), see Eq. (A23). Despite the messiness of this solution, only one unknown can be solved for, because the entire problem reduces to satisfying Eq. (A13). If R_1 and $R_{\rm tamp}$ are specified, then one can solve numerically for R_2. The masses and radii are then related by

$$R_1 = \left(\frac{3M_1}{4\pi\rho_1}\right)^{1/3}, \tag{A24}$$

$$R_2 = \left[\frac{3}{4\pi}\left(\frac{M_1}{\rho_1} + \frac{M_2}{\rho_2}\right)\right]^{1/3}, \tag{A25}$$

and

$$R_{\rm tamp} = \left[\frac{3}{4\pi}\left(\frac{M_1}{\rho_1} + \frac{M_2}{\rho_2} + \frac{M_{\rm tamp}}{\rho_{\rm tamp}}\right)\right]^{1/3}. \tag{A26}$$

Any compression involved can be accounted for by multiplying the densities by the desired compression ratio.

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Demonstration Screw

The presence of this model screw in the apparatus collection at the University of Utah might seem trivial, but it has historical precedence, for all of the nineteenth century physics apparatus catalogues show it. Today there are probably many students who are so removed from our rural past that they are unfamiliar with this vital piece of mechanism. Perhaps some advanced students do not know the geometrical meaning of a screw dislocation in a crystal. This example is listed at $4.00 in the 1888 catalogue of James W. Queen of Philadelphia. (Picture and text by Thomas B. Greenslade, Jr., Kenyon College)